Temporally-structured multimodal prediction across six clinical targets in paediatric acute lymphoblastic leukaemia

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Abstract

Paediatric acute lymphoblastic leukaemia (ALL) is the most common childhood malignancy, yet 15–20% of patients relapse after apparently successful treatment. Existing risk-stratification approaches operate on single-timepoint snapshots and have not been systematically evaluated across the full clinical decision timeline nor validated across multiple independent centres. Here we present a temporally-structured multimodal prediction framework evaluated on six clinical targets simultaneously: four classification tasks (Final Risk, D19MRD, D46MRD, Relapse) and two survival endpoints (event-free survival, EFS; overall survival, OS), whose feature sets are strictly aligned with the clinical data-acquisition timeline from hospital admission to Day 46. An automated morphology pipeline converts bone marrow smear images into structured cell-level profiles, and we systematically test—at every feature level—whether raw images or derived morphology add predictive signal beyond the clinical features—signal that current risk-stratification rules leave unused. Phase 1 evaluates four large language models (LLMs) in a zero-shot setting; Phase 2 trains decision trees, random forests, and random survival forests on a temporally partitioned case database. In the internal cohort, Phase 2 achieves AUROC 0.949 for Final Risk and concordance index 0.746 for EFS. A cosine similarity retrieval-augmented generation (RAG) belief revision improves D46MRD AUROC from 0.655 to 0.881 (**+0.226**) in the largest training regime. External validation on a merged 41-patient cohort pooled from two hospitals confirms robust Final Risk transfer (LLM zero-shot AUROC up to 0.925; Phase 2 RF up to 0.991) without retraining; pooling also makes Relapse and D19MRD two-class and AUROC-evaluable,

although the cohort remains small and, because one source hospital is a relapse-enriched referral series, external relapse metrics are exploratory. This framework demonstrates that temporally stratified multimodal learning, aligned with clinical workflows, enables consistent multi-target prediction that generalises to independent cohorts.

Keywords: paediatric acute lymphoblastic leukaemia, temporal feature design, large language models, risk stratification, survival analysis, retrieval-augmented generation, external validation

1 Introduction

Acute lymphoblastic leukaemia (ALL) is the most common childhood malignancy, accounting for approximately 25% of all paediatric cancers [1]. Risk-adapted chemotherapy has elevated five-year event-free survival (EFS) rates above 90% in high-income settings, yet the 15–20% of patients who relapse face substantially worse outcomes, with long-term survival dropping to 30–50% upon re-induction [2, 3]. Because the majority of relapses occur after apparent clinical remission, the prospective identification of high-risk patients—early enough to intensify or de-escalate therapy—remains a central and unresolved challenge in paediatric oncology.

Current standard-of-care risk stratification—Initial Risk at diagnosis and Final Risk after the Day 46 induction assessment—relies on pre-specified categorical cut-offs for age, white blood cell count, cytogenetics, immunophenotype, and minimal residual disease (MRD). These rules do not fully exploit the multimodal information that accumulates during treatment, nor do they yield personalised probability estimates for future relapse or survival. Computational approaches promise to fill this gap, but most published models make simplifications that break down at the bedside: they treat outcome prediction as a static, single-shot problem, when in reality paediatric ALL management unfolds as a structured sequence of events in time.

We argue that realistic clinical prediction is governed by three distinct and largely independent timelines, and that existing methods collapse or ignore all three. The first is the *feature-acquisition timeline*: a patient's data do not arrive at once but in a fixed clinical order—white blood cell count and age at admission (Day 0), immunophenotype (Day 1), fusion genes, karyotype and FISH (Day 7–10), and MRD (Day 19–46). A model trained on features pooled across all timepoints implicitly uses information unavailable at the moment of decision, producing optimistic but non-deployable estimates [9]. The second is the *prediction-target timeline*: clinicians do not make one prediction but a sequence of decisions—early response (D19MRD), end-of-induction response (D46MRD), Final Risk assignment, and long-horizon outcomes (relapse, EFS, overall survival [OS]). Studying any single target in isolation, as is common, obscures how one data-acquisition workflow must support all six decision nodes. The third is the *data-accumulation timeline*: a hospital's reference database grows year by year, so a model deployed in a given year can only have learned from patients enrolled before it. Random shuffled train–test splits violate this constraint and report performance that

is not prospectively attainable. Aligning a dataset and a method to all three timelines simultaneously, rather than one at a time, is the central novelty of this work and the source of its clinical realism.

Building on this principle, we construct a dataset in which seven temporally ordered clinical feature sets (six strictly nested; $\mathcal{F}_4$ is a Day-1 branch of $\mathcal{F}_3$) map onto the feature-acquisition timeline, six prediction targets map onto the target timeline, and patient enrolment year defines the data-accumulation timeline. On this substrate we evaluate two complementary methods. Phase 1 applies large language models (LLMs) in a zero-shot setting, probing the predictive upper bound reachable with *no* accumulated training data—the regime a clinician faces when a new protocol or a new hospital comes online [10]. Phase 2 trains decision trees, random forests, and random survival forests on the growing case database under temporal splits, emulating prospective deployment as the data-accumulation timeline advances. A retrieval-augmented generation (RAG) component then leverages the accumulated database to apply a similarity-weighted belief revision to model probabilities and to generate interpretable clinical narratives. In parallel, an automated bone marrow morphology pipeline (SAM-based cell segmentation and BiomedCLIP clustering) converts smear images into a structured cell-morphology profile, letting us test—at every level of the clinical feature hierarchy—whether imaging adds prognostic information that current risk rules and prior computational studies leave unused.

Our contributions are: (*i*) a seven-level, three-timeline clinical feature-and-target hierarchy aligned with the clinical workflow; (*ii*) Phase 1, a systematic zero-shot evaluation of four LLMs across all feature levels and all six prediction targets, establishing training-free baselines; (*iii*) a systematic test of whether bone marrow smear images—unused by current risk-stratification rules—add predictive signal beyond clinical features, finding that a structured morphology profile derived from the images carries relapse-relevant signal at diagnosis; (*iv*) Phase 2, a decision tree, random forest, and random survival forest system trained on a dynamic case database partitioned by enrolment year and validated under three temporal splits; (*v*) a cosine-similarity RAG belief revision that refines model probabilities and generates interpretable LLM clinical narratives; and (*vi*) multi-site external validation at independent centres that received no training data.

2 Results

2.1 Study cohorts and temporally-structured feature design

The internal training cohort comprised paediatric ALL patients treated at [HOSPITAL NAME] between 2015 and 2019 (Table 1; $n = 301$). Prediction targets span the treatment timeline: Day 19 MRD positivity (D19MRD $\geq 1\%$), Day 46 MRD positivity (D46MRD $\geq 0.01\%$), final risk classification (Final Risk: LR vs IR/HR; 159 LR, 142 IR/HR), and relapse occurrence (44 of 301, 14.6%). EFS events (relapse, induction failure, or death; 75 of 301, 24.9%) and OS events (death; 24 of 301, 8.0%) together extend the prediction targets to six.

External validation used a single pooled cohort of 41 patients collected at two hospitals: Lianyungang First People's Hospital (20 patients, diagnosed 2019–2023) and

Children's Hospital of Fudan University, Shanghai (21 patients, 2017–2021) (Table 1). The Fudan sub-cohort is a relapse-enriched referral series (all 21 patients relapsed), so the pooled external relapse rate (23 of 41, 56.1%) sits far above that of an unselected ALL population; external relapse metrics are therefore exploratory (see Limitations). Pooling the two hospitals resolves the single-class collapse a per-site split would otherwise cause for several targets: D19MRD and Relapse become two-class and AUROC-evaluable across the full cohort, while D46MRD still has only one positive patient after pooling and is reported as accuracy. For external validation, all 301 internal patients were used as the training set and no external data were used in training.

Feature sets $\mathcal{F}_1$–$\mathcal{F}_7$ (Table 13) are temporally ordered and, except for the $\mathcal{F}_4$ branch, strictly nested: each level adds only features available at the corresponding clinical timepoint. $\mathcal{F}_1$ contains WBC count and age (Day 0); $\mathcal{F}_7$ additionally includes D19MRD and D46MRD values (Day 46). The feature hierarchy applies identically to all six prediction targets, with the distinction that classification tasks (targets 1–4) use AUROC as the primary metric, while survival targets (targets 5–6: EFS, OS) use Harrell's C-index and time-dependent IPCW AUC.

Table 1 Cohort characteristics and prediction target distributions. The External cohort pools patients from two independent hospitals (see Methods for site details). † patients with complete labels for all four classification targets. ‡ patients whose automated pipeline output contains at least one blast cell; only these enter the imaging analysis (Section 2.3).

Characteristic	**Internal** ($n = 301$)	**External** ($n = 41$)
Enrolment years	2015–2019	2017–2023
B-ALL	163 (54.2%)	38 (92.7%)
Patients with complete labels†	301	29
Classification targets		
Final Risk: LR	159 (52.8%)	17 (41.5%)
Final Risk: IR/HR	142 (47.2%)	24 (58.5%)
Relapse: Yes	44 (14.6%)	23 (56.1%)
D19MRD $\geq 1\%$	[X] ([X]%)	4
D46MRD $\geq 0.01\%$	[X] ([X]%)	1
Survival endpoints		
EFS events	75 (24.9%)	24 (58.5%)
OS events	24 (8.0%)	7 (17.1%)
Multimodal data		
Bone marrow images	[X] (XX%)	41
Morphology features	287 (95.3%)	40
non-empty blast profile‡	221 (73.4%)	13

Table 2 Phase 1 LLM zero-shot performance across all seven clinical feature levels and six prediction targets (internal cohort, $n = 301$). Each cell is the macro-averaged AUROC (classification targets) or Harrell's concordance index (survival targets) for one LLM at one feature level; **bold** marks each model's best feature level for that target (column maximum). Classification AUROC is computed from parsed categorical predictions (balanced-accuracy form). — indicates a target not evaluated at that level: D19/D46MRD are used as *input* features at $\mathcal{F}_7$ (see Table 13).

	Model					Model			
Lv.	Gem.	Gma.	Int.	Qw.	**Lv.**	Gem.	Gma.	Int.	Qw.
Final Risk (AUROC)					*Relapse (AUROC)*				
$\mathcal{F}_1$	0.788	0.785	0.729	0.792	$\mathcal{F}_1$	0.514	0.572	0.566	0.530
$\mathcal{F}_2$	0.782	0.796	0.539	0.789	$\mathcal{F}_2$	0.557	0.564	0.519	0.526
$\mathcal{F}_3$	0.812	0.842	0.583	0.831	$\mathcal{F}_3$	0.577	0.572	0.503	0.503
$\mathcal{F}_4$	0.826	0.835	0.591	0.837	$\mathcal{F}_4$	0.575	0.555	0.500	0.552
$\mathcal{F}_5$	0.791	0.854	0.823	0.845	$\mathcal{F}_5$	0.596	0.558	0.554	0.504
$\mathcal{F}_6$	0.834	**0.891**	**0.848**	**0.882**	$\mathcal{F}_6$	**0.599**	0.552	**0.643**	0.546
$\mathcal{F}_7$	**0.849**	0.788	0.716	0.856	$\mathcal{F}_7$	0.567	**0.586**	0.540	**0.557**
D19 MRD (AUROC)					*EFS (C-index)*				
$\mathcal{F}_1$	0.571	0.543	0.564	0.531	$\mathcal{F}_1$	0.577	0.591	0.584	0.573
$\mathcal{F}_2$	0.543	0.555	0.479	0.521	$\mathcal{F}_2$	0.607	0.598	0.522	0.561
$\mathcal{F}_3$	0.631	0.599	0.484	0.557	$\mathcal{F}_3$	0.587	0.614	0.545	0.537
$\mathcal{F}_4$	0.583	0.597	0.488	0.520	$\mathcal{F}_4$	0.598	0.614	0.526	0.561
$\mathcal{F}_5$	0.603	0.620	**0.567**	**0.605**	$\mathcal{F}_5$	0.624	0.583	0.604	0.542
$\mathcal{F}_6$	**0.651**	**0.652**	0.548	0.581	$\mathcal{F}_6$	0.643	0.630	0.621	0.581
$\mathcal{F}_7$	—	—	—	—	$\mathcal{F}_7$	**0.687**	**0.716**	**0.642**	**0.662**
D46 MRD (AUROC)					*OS (C-index)*				
$\mathcal{F}_1$	0.505	0.486	**0.542**	0.501	$\mathcal{F}_1$	0.689	0.648	0.652	0.661
$\mathcal{F}_2$	0.492	0.475	0.501	0.495	$\mathcal{F}_2$	0.647	0.702	0.550	0.634
$\mathcal{F}_3$	0.502	0.509	0.503	0.500	$\mathcal{F}_3$	0.682	0.678	0.537	0.634
$\mathcal{F}_4$	0.505	0.539	0.505	0.533	$\mathcal{F}_4$	0.687	0.691	0.497	0.660
$\mathcal{F}_5$	0.549	**0.577**	0.518	**0.552**	$\mathcal{F}_5$	0.701	0.704	0.734	0.619
$\mathcal{F}_6$	**0.610**	0.539	0.523	0.510	$\mathcal{F}_6$	0.667	0.693	0.755	0.687
$\mathcal{F}_7$	—	—	—	—	$\mathcal{F}_7$	**0.725**	**0.718**	**0.764**	**0.718**

2.2 Phase 1: LLM zero-shot performance across six targets

We evaluated four LLMs in a zero-shot setting across all seven clinical feature levels and all six prediction targets; Table 2 reports the complete internal per-feature-level results and Table 3 the merged external cohort. Models included Gemini-3.1-Flash-Lite (Google AI API) and three open-weight models deployed locally via vLLM: Gemma-4-31B-it, InternVL3.5-30B-A3B, and Qwen3.6-35B-A3B. For classification tasks (targets 1–4), each patient's feature vector was serialised into natural language and the model output a categorical label; we report AUROC. For survival targets (5–6), the model output a continuous risk score (0–1) evaluated by concordance index (C-index).

Classification performance on the internal cohort. Final Risk was the easiest target for all models. At Day 0 features ($\mathcal{F}_1$: WBC and age), all four models already achieved AUROC $\approx$ 0.73–0.79, reflecting the known prognostic importance of WBC count. Performance increased incrementally and peaked at the cytogenetics panel $\mathcal{F}_6$ (Gemini 0.834, Gemma 0.891, InternVL 0.848, Qwen 0.882), which was the best level for the three open-weight models; supplying the D19/D46 MRD values as inputs at $\mathcal{F}_7$ raised Gemini to 0.849 but did not further improve the others. Best-level Final Risk AUROCs were thus 0.849 (Gemini, $\mathcal{F}_7$), 0.891 (Gemma, $\mathcal{F}_6$), 0.848 (InternVL, $\mathcal{F}_6$), and 0.882 (Qwen, $\mathcal{F}_6$). Relapse prediction was substantially harder: best AUROCs were 0.643 (InternVL, $\mathcal{F}_6$), 0.599 (Gemini, $\mathcal{F}_6$), 0.586 (Gemma, $\mathcal{F}_7$), and 0.557 (Qwen, $\mathcal{F}_7$), with no consistent improvement as features accumulated. D19MRD and D46MRD showed moderate performance (AUROC 0.55–0.65), suggesting that LLMs can partially infer early treatment response from immunophenotypic and cytogenetic profiles but that these tasks remain harder without direct MRD measurement.

Survival performance on the internal cohort. For EFS, best-level C-indices were all attained at $\mathcal{F}_7$: 0.716 (Gemma), 0.687 (Gemini), 0.662 (Qwen), and 0.642 (InternVL). The inclusion of the D19/D46 MRD values at $\mathcal{F}_7$ was consistently optimal for EFS, confirming that residual disease burden encodes strong prognostic information accessible to LLMs without labelled training data. For OS—with only 24 events—C-indices also peaked at $\mathcal{F}_7$: InternVL reached 0.764, Gemini 0.725, and Gemma and Qwen both 0.718. Log-rank tests were significant ($p < 0.05$) at $\mathcal{F}_7$ for EFS for all four LLMs.

External validation (merged cohort). Final Risk transferred well on the pooled 41-patient external cohort: LLM zero-shot AUROC ranged 0.450–0.925 across models and feature levels, peaking at $\mathcal{F}_7$ (0.750–0.925 across the four LLMs), close to the internal range and obtained without any cohort-specific retraining (Table 3). Pooling the two source hospitals resolves the single-class collapse that a per-site split would otherwise produce for Relapse and D19MRD: both targets are now two-class across the full external cohort and AUROC-evaluable (Relapse 0.482–0.717; D19MRD close to chance, 0.340–0.550); only D46MRD still has a single positive patient after pooling (1 of 29 labelled patients) and is reported with accuracy (0.035–0.724) rather than AUROC. For survival, the pooled cohort (24 EFS events, 7 deaths) gave LLM EFS C-indices of 0.542–0.715 and OS 0.548–0.859.

2.3 Testing whether bone marrow imaging adds prognostic information beyond clinical features

Bone marrow smear microscopy is performed for every ALL patient at diagnosis, yet current risk-stratification rules use only the blast percentage derived from it—the images themselves enter no downstream risk calculation, and prior computational studies of paediatric ALL risk have likewise not tested whether the images carry additional prognostic signal. We therefore evaluate, at every clinical feature level $\mathcal{F}_3$–$\mathcal{F}_7$, three ways of injecting imaging information into the zero-shot LLMs: (+img) the raw smear images as direct multimodal input; (+morph) a structured morphology profile generated by a four-stage automated pipeline (SAM segmentation, $\sim$37 morphological features per cell, BiomedCLIP cell-type clustering, per-patient aggregation;

Table 3 Phase 1 LLM zero-shot on the pooled external cohort, all feature levels (Gm/Gma/Iv/Qw = Gemini/Gemma/InternVL/Qwen). The two source hospitals are pooled into a single external cohort (n=41 for survival; classification targets use the n=29 subset with available labels). Classification: AUROC (* = accuracy; D46MRD has only one positive class after pooling, so accuracy is reported instead). Survival (EFS, OS): C-index. **Bold**: best feature level per column within each task block (ties all bolded). —: not evaluable ($\mathcal{F}_7$ not defined for D19/D46 MRD). External pooled cohort; exploratory, indicative only.

Lv.	Gm	Gma	Iv	Qw
Final Risk (AUROC)				
$\mathcal{F}_1$	0.739	0.794	0.714	0.794
$\mathcal{F}_2$	0.794	0.819	0.450	0.794
$\mathcal{F}_3$	0.708	0.819	0.531	0.794
$\mathcal{F}_4$	0.844	0.819	0.697	0.794
$\mathcal{F}_5$	0.653	**0.894**	0.536	0.739
$\mathcal{F}_6$	0.708	0.844	0.733	0.819
$\mathcal{F}_7$	**0.925**	0.775	**0.750**	**0.850**
D19 MRD (AUROC)				
$\mathcal{F}_1$	0.425	0.425	0.490	0.380
$\mathcal{F}_2$	0.530	0.405	0.500	**0.550**
$\mathcal{F}_3$	0.425	0.405	0.500	0.340
$\mathcal{F}_4$	**0.550**	0.405	0.500	0.445
$\mathcal{F}_5$	0.425	**0.465**	0.395	0.445
$\mathcal{F}_6$	0.425	0.405	**0.515**	0.545
$\mathcal{F}_7$	—	—	—	—
D46 MRD (accuracy; single positive)				
$\mathcal{F}_1$	**0.690***	0.586*	**0.586***	0.621*
$\mathcal{F}_2$	0.586*	0.552*	0.035*	0.655*
$\mathcal{F}_3$	0.586*	0.586*	0.035*	0.690*
$\mathcal{F}_4$	0.552*	0.586*	0.035*	**0.724***
$\mathcal{F}_5$	0.586*	**0.621***	0.138*	0.690*
$\mathcal{F}_6$	0.621*	0.586*	0.207*	0.655*
$\mathcal{F}_7$	—	—	—	—
Relapse (AUROC)				
$\mathcal{F}_1$	0.587	0.634	**0.699**	0.482
$\mathcal{F}_2$	0.504	0.612	0.500	0.482
$\mathcal{F}_3$	0.674	0.551	0.562	0.544
$\mathcal{F}_4$	0.609	0.551	0.685	0.565
$\mathcal{F}_5$	**0.717**	0.612	0.620	0.587
$\mathcal{F}_6$	0.634	**0.656**	0.536	**0.609**
$\mathcal{F}_7$	0.500	0.522	0.500	0.522
EFS (C-index)				
$\mathcal{F}_1$	0.657	**0.669**	0.627	0.634
$\mathcal{F}_2$	**0.715**	0.659	0.551	**0.645**
$\mathcal{F}_3$	0.635	0.590	0.615	0.638
$\mathcal{F}_4$	0.634	0.590	0.615	0.638
$\mathcal{F}_5$	0.617	0.656	0.639	0.542
$\mathcal{F}_6$	0.657	0.651	**0.709**	0.567
$\mathcal{F}_7$	0.656	0.664	0.702	0.591
OS (C-index)				
$\mathcal{F}_1$	0.808	**0.821**	0.802	0.714
$\mathcal{F}_2$	0.804	0.792	0.633	0.744
$\mathcal{F}_3$	0.746	0.780	0.710	0.548
$\mathcal{F}_4$	0.760	0.780	0.710	0.548
$\mathcal{F}_5$	0.762	0.808	0.816	0.694
$\mathcal{F}_6$	**0.810**	0.603	0.814	**0.806**
$\mathcal{F}_7$	0.802	0.615	**0.859**	0.665

Methods, Section 4.3), serialised as text; (+both) images and profile together. To keep comparisons fair, each level's base row is re-evaluated on the same patient subset as its imaging variants (patients with available morphology data). Table 4 reports the internal cohort; Tables 5 and 6 report the merged external cohort.

With the comparison now complete at all five levels, the clearest and most consistent internal-cohort gain is for relapse prediction from the morphology profile: +morph improves Relapse AUROC over base for a majority of models (3–4 of 4) at three of the five levels (4/4 at $\mathcal{F}_3$ and $\mathcal{F}_5$, 3/4 at $\mathcal{F}_6$), with per-model deltas ranging from -0.043 to $+0.070$ across levels. Raw images help less reliably: +img improves Final Risk for 3/4 models at $\mathcal{F}_3$ and $\mathcal{F}_6$ but for at most 2/4 at the other levels, and improves Relapse for only 1/4 models at $\mathcal{F}_3$–$\mathcal{F}_5$ (4/4 at $\mathcal{F}_6$). At $\mathcal{F}_7$, where the D19/D46 MRD measurements are already inputs, the morphology profile still adds Final Risk signal—improving all four models ($+0.005$ to $+0.102$)—whereas raw images degrade Final Risk for all four (-0.034 to -0.005); Relapse changes at $\mathcal{F}_7$ are small for every variant (-0.019 to $+0.027$). For survival, +morph is again the only variant with a consistent benefit: it improves EFS over base in 15 of 20 model–level comparisons (4/4 models at $\mathcal{F}_4$, 3/4 at $\mathcal{F}_5$–$\mathcal{F}_7$; deltas -0.027 to $+0.033$), whereas +img (9/20) and +both (6/20) are net neutral to harmful, and for OS no variant helps consistently (at best +morph, 8/20; +img degrades OS by up to -0.235). The morphology profile also strengthens the tabular Phase 2 models: adding the quantitative morphology features to $\mathcal{F}_3$ raised D19MRD to DT AUROC 0.743 / RF AUROC 0.807 ($\mathcal{F}_3^{+\mathrm{m}}$, Table 8), showing that quantitative cell-level characterisation carries prognostic signal available already at diagnosis. These are unadjusted differences on a single internal cohort, without significance testing, and should be read as directional evidence; the external fair-comparison subsets (Tables 5 and 6) are too small to confirm or refute them.

2.4 Phase 2: Temporal validation and classification+survival results

Phase 2 evaluates DT and RF models on three temporal splits ($\mathcal{T}_1 = 2016$: $n_{\text{train}} = 99$, $n_{\text{test}} = 201$; $\mathcal{T}_2 = 2017$: $n_{\text{train}} = 186$, $n_{\text{test}} = 114$; $\mathcal{T}_3 = 2018$: $n_{\text{train}} = 283$, $n_{\text{test}} = 17$) and a random survival forest (RSF) on the same splits for the EFS and OS targets (internal results in Table 7, external in Table 9).

Classification on the internal cohort (Split 2). Final Risk DT AUROC rose from 0.810 at $\mathcal{F}_1$ to 0.912 at $\mathcal{F}_6$, with the largest single gain at $\mathcal{F}_5$ ($+0.127$, introduction of Initial Risk and Day 5 blast percentage). Including MRD at $\mathcal{F}_7$ further raised DT AUROC to **0.949** (RF: 0.998), the highest classification result across all conditions. D19MRD achieved DT AUROC 0.743 and RF AUROC 0.807 at $\mathcal{F}_3^{+\mathrm{morph}}$, where the quantitative cell-morphology features from the pipeline augment the tabular model; D46MRD reached DT 0.671 and RF 0.636 at $\mathcal{F}_4$. Relapse prediction remained difficult for Phase 2 as for Phase 1 (best DT AUROC 0.529, RF 0.630), mirroring the broader literature on relapse biology [8].

Survival on the internal cohort (Split 2). RSF at $\mathcal{F}_7$ achieved C-index **0.746** and time-AUC at 24 months 0.775 for EFS (log-rank $p = 0.010$), outperforming the best LLM zero-shot EFS C-index (0.716) obtained without any training data. For

Table 4 Information gain from bone marrow imaging at every clinical feature level (internal cohort). For each base level $\mathcal{F}_k$ ($k = 3..7$): +img adds raw smear images, +morph the structured morphology profile, +both adds both. Classification targets: AUROC; EFS/OS: C-index. Base rows are re-evaluated on the subset of patients with a non-empty automated morphology profile (blast cells detected by the pipeline), so they can differ slightly from Table 2. **Bold**: best variant within each level block (per model column; ties all bolded). —: not applicable (D19/D46MRD are inputs at $\mathcal{F}_7$).

	Gm	Gma	Iv	Qw		Gm	Gma	Iv	Qw
Final Risk (AUROC)					*Relapse (AUROC)*				
$\mathcal{F}_3$	0.792	0.827	0.558	0.812	$\mathcal{F}_3$	0.543	0.532	0.505	0.481
+img	**0.801**	0.823	**0.561**	**0.822**	+img	0.484	0.513	0.484	0.486
+morph	0.759	**0.833**	0.546	0.798	+morph	**0.565**	**0.535**	**0.511**	**0.521**
+both	0.767	0.811	0.552	0.769	+both	0.554	0.497	0.503	0.510
$\mathcal{F}_4$	**0.811**	0.822	0.560	0.817	$\mathcal{F}_4$	0.540	0.511	0.484	**0.536**
+img	0.757	**0.823**	0.579	0.809	+img	0.508	**0.524**	0.465	0.490
+morph	0.782	0.821	**0.616**	**0.839**	+morph	0.557	0.468	0.511	0.515
+both	0.776	0.794	0.577	0.800	+both	**0.562**	0.508	**0.516**	0.508
$\mathcal{F}_5$	0.783	0.839	0.804	0.827	$\mathcal{F}_5$	0.602	0.546	0.540	0.502
+img	0.763	0.806	**0.840**	0.812	+img	0.586	0.513	0.557	0.491
+morph	0.784	0.779	0.795	**0.843**	+morph	**0.616**	**0.594**	**0.581**	0.550
+both	**0.788**	**0.844**	0.775	0.795	+both	**0.616**	0.551	0.530	**0.572**
$\mathcal{F}_6$	0.818	0.872	0.832	0.874	$\mathcal{F}_6$	0.546	0.479	0.618	0.486
+img	**0.840**	0.880	**0.838**	0.873	+img	**0.624**	0.495	**0.621**	**0.505**
+morph	0.820	0.875	0.800	**0.876**	+morph	0.616	**0.527**	0.602	0.492
+both	0.821	**0.881**	0.822	0.862	+both	0.600	0.508	0.565	0.502
$\mathcal{F}_7$	0.837	0.761	0.699	0.852	$\mathcal{F}_7$	**0.589**	0.591	0.559	0.564
+img	0.803	0.745	0.694	0.832	+img	**0.589**	**0.597**	0.567	**0.578**
+morph	**0.842**	**0.770**	**0.801**	**0.858**	+morph	**0.589**	0.573	0.557	0.564
+both	0.786	0.731	0.715	0.765	+both	0.575	0.581	**0.586**	**0.578**
Avg	**0.808**	**0.824**	0.691	0.836	Avg	0.564	0.532	0.541	0.514
+img	0.793	0.815	0.702	0.830	+img	0.558	0.528	0.539	0.510
+morph	0.797	0.816	**0.712**	**0.843**	+morph	**0.588**	**0.539**	**0.552**	0.529
+both	0.787	0.812	0.688	0.798	+both	0.581	0.529	0.540	**0.534**
D19 MRD (AUROC)					*EFS (C-index)*				
$\mathcal{F}_3$	**0.615**	0.608	0.478	0.575	$\mathcal{F}_3$	0.550	0.585	**0.513**	0.532
+img	0.590	**0.704**	0.499	**0.644**	+img	0.580	0.575	0.448	**0.552**
+morph	0.538	0.677	0.488	0.412	+morph	0.567	**0.593**	0.496	0.525
+both	0.579	0.643	**0.512**	0.461	+both	**0.602**	0.576	0.452	0.522
$\mathcal{F}_4$	0.591	0.606	0.478	0.519	$\mathcal{F}_4$	0.559	0.584	0.484	0.555
+img	**0.626**	0.642	**0.508**	0.548	+img	**0.612**	**0.600**	0.456	**0.572**
+morph	0.609	**0.658**	0.493	**0.613**	+morph	0.568	0.585	**0.502**	0.560
+both	0.584	0.586	**0.508**	0.536	+both	0.577	0.589	0.470	0.547
$\mathcal{F}_5$	0.612	0.632	**0.574**	**0.644**	$\mathcal{F}_5$	0.608	0.552	**0.574**	0.520
+img	0.615	0.636	0.504	0.606	+img	0.585	0.577	0.473	0.500
+morph	**0.655**	0.608	0.538	0.611	+morph	**0.609**	**0.585**	0.574	**0.535**
+both	0.644	**0.655**	0.501	0.589	+both	0.584	0.582	0.570	0.503
$\mathcal{F}_6$	0.678	0.669	0.560	0.606	$\mathcal{F}_6$	0.625	**0.613**	0.606	0.547
+img	0.660	**0.694**	**0.590**	**0.679**	+img	0.602	0.598	0.556	0.550
+morph	0.636	0.684	0.556	0.591	+morph	**0.650**	0.586	**0.610**	0.577
+both	**0.720**	0.684	0.563	0.627	+both	0.624	0.585	0.578	**0.593**
$\mathcal{F}_7$	—	—	—	—	$\mathcal{F}_7$	0.658	0.700	0.615	0.622
+img	—	—	—	—	+img	0.659	0.668	0.518	**0.631**
+morph	—	—	—	—	+morph	**0.667**	**0.701**	**0.636**	0.621
+both	—	—	—	—	+both	0.666	0.650	0.545	0.599
Avg	0.624	0.629	0.522	0.586	Avg	0.600	0.607	0.558	0.555
+img	0.623	**0.669**	**0.525**	**0.619**	+img	0.608	0.604	0.490	0.561
+morph	0.609	0.657	0.519	0.557	+morph	**0.612**	**0.610**	**0.563**	**0.564**
+both	**0.632**	0.642	0.521	0.553	+both	0.610	0.596	0.523	0.553
D46 MRD (AUROC)					*OS (C-index)*				
$\mathcal{F}_3$	0.501	0.488	0.498	0.488	$\mathcal{F}_3$	0.684	0.630	**0.482**	0.622
+img	0.516	0.524	**0.506**	0.498	+img	0.672	0.639	0.343	0.584
+morph	0.535	0.505	0.505	0.519	+morph	0.634	0.633	0.460	0.595
+both	**0.560**	**0.589**	0.420	**0.547**	+both	**0.687**	**0.647**	0.430	**0.637**
$\mathcal{F}_4$	**0.509**	0.529	0.495	**0.558**	$\mathcal{F}_4$	0.705	**0.663**	**0.439**	0.639
+img	0.501	0.551	0.508	0.546	+img	0.672	0.635	0.387	0.576
+morph	0.505	0.546	0.477	0.485	+morph	0.660	0.652	0.430	**0.677**
+both	0.502	**0.582**	**0.521**	0.493	+both	**0.714**	0.649	0.366	0.600
$\mathcal{F}_5$	0.525	0.565	0.494	0.560	$\mathcal{F}_5$	0.695	**0.701**	**0.720**	0.602
+img	**0.626**	0.504	0.529	0.496	+img	0.682	0.622	0.485	0.509
+morph	0.598	0.508	**0.542**	0.488	+morph	**0.728**	0.670	0.712	**0.620**
+both	0.578	**0.570**	0.466	**0.572**	+both	0.683	0.602	0.691	0.530
$\mathcal{F}_6$	0.606	0.524	**0.536**	0.503	$\mathcal{F}_6$	0.658	0.683	**0.742**	**0.665**
+img	0.604	0.565	0.499	0.497	+img	0.656	0.633	0.580	0.588
+morph	**0.685**	**0.592**	0.487	**0.528**	+morph	0.675	**0.701**	0.704	0.638
+both	0.679	0.531	0.511	0.471	+both	**0.680**	0.658	0.680	0.583
$\mathcal{F}_7$	—	—	—	—	$\mathcal{F}_7$	0.680	0.698	**0.752**	0.668
+img	—	—	—	—	+img	0.669	**0.718**	0.532	0.677
+morph	—	—	—	—	+morph	**0.702**	0.693	0.739	**0.685**
+both	—	—	—	—	+both	0.699	0.694	0.639	0.678
Avg	0.535	0.526	0.506	**0.527**	Avg	0.684	**0.675**	**0.627**	0.639
+img	0.561	0.536	**0.510**	0.509	+img	0.670	0.649	0.465	0.587
+morph	**0.581**	0.538	0.503	0.505	+morph	0.680	0.670	0.609	**0.643**
+both	0.580	**0.568**	0.479	0.521	+both	**0.692**	0.650	0.561	0.605

Table 5 Information gain from bone marrow imaging at every clinical feature level, external classification tasks (merged external cohort). Gm/Gma/Iv/Qw = Gemini/Gemma/InternVL/Qwen. For each base level $\mathcal{F}_k$ ($k = 3..7$ for Final Risk and Relapse, $k = 3..6$ for D19/D46MRD): +img adds raw smear images, +morph the structured morphology profile, +both adds both. AUROC (* = accuracy, single-class task/level). Base rows are re-evaluated on the morphology-available patient subset, so they can differ slightly from Table 3. **Bold**: best variant within each level block (per column; ties all bolded). The pooled external fair-comparison subset is small (n=8; one model–level cell n=7 owing to a parse-filtered response); indicative only.

	Gm	Gma	Iv	Qw
*Final Risk (AUROC; * =accuracy)*				
$\mathcal{F}_3$	**0.750**	0.750	**0.625**	0.750
+img	**0.750**	0.750	0.500	0.750
+morph	0.625	0.750	**0.625**	**0.875**
+both	**0.750**	**0.875**	**0.625**	0.500
$\mathcal{F}_4$	**0.750**	**0.750**	**0.750**	**0.750**
+img	**0.750**	**0.750**	0.625	**0.750**
+morph	**0.750**	**0.750**	**0.750**	**0.750**
+both	0.625	**0.750**	0.625	0.625
$\mathcal{F}_5$	**0.875**	**0.875**	0.625	0.750
+img	0.750	**0.875**	**0.750**	0.750
+morph	0.750	0.750	**0.750**	**0.875**
+both	**0.875**	**0.875**	**0.750**	**0.875**
$\mathcal{F}_6$	**0.875**	**0.875**	**0.875**	**0.875**
+img	**0.875**	**0.875**	**0.875**	**0.875**
+morph	**0.875**	**0.875**	**0.875**	**0.875**
+both	**0.875**	**0.875**	**0.875**	**0.875**
$\mathcal{F}_7$	**1.000**	0.750	0.750	**0.875**
+img	**1.000**	**0.875**	**0.875**	0.750
+morph	**1.000**	0.625	0.750	**0.875**
+both	0.750	0.750	0.625	**0.875**
Avg	**0.850**	0.800	0.725	0.800
+img	0.825	**0.825**	0.725	0.775
+morph	0.800	0.750	**0.750**	**0.850**
+both	0.775	**0.825**	0.700	0.750
*D19 MRD (AUROC; * =accuracy)*				
$\mathcal{F}_3$	**0.625***	**0.625***	0.000*	**0.625***
+img	**0.625***	0.500*	0.000*	0.500*
+morph	0.375*	0.125*	0.000*	**0.625***
+both	0.375*	0.000*	**0.125***	0.375*
$\mathcal{F}_4$	**0.625***	**0.625***	**0.000***	**0.625***
+img	**0.625***	0.500*	**0.000***	0.250*
+morph	0.375*	0.125*	**0.000***	0.500*
+both	0.375*	0.125*	**0.000***	0.375*
$\mathcal{F}_5$	**0.875***	**0.750***	0.000*	**0.750***
+img	**0.875***	**0.750***	0.000*	**0.750***
+morph	**0.875***	0.375*	**0.250***	**0.750***
+both	0.750*	0.375*	0.000*	**0.750***
$\mathcal{F}_6$	**0.750***	**0.500***	0.375*	**1.000***
+img	**0.750***	**0.500***	**0.500***	0.500*
+morph	0.625*	0.375*	0.250*	0.750*
+both	0.625*	**0.500***	0.250*	0.875*
Avg	**0.719***	**0.625***	0.094*	**0.750***
+img	**0.719***	0.563*	**0.125***	0.500*
+morph	0.563*	0.250*	**0.125***	0.656*
+both	0.531*	0.250*	0.094*	0.594*
D46 MRD (accuracy; single positive)				
$\mathcal{F}_3$	**0.625***	**0.625***	0.000*	**0.625***
+img	**0.625***	0.500*	**0.125***	0.500*
+morph	0.500*	0.125*	0.000*	0.250*
+both	0.500*	0.125*	**0.125***	0.375*
$\mathcal{F}_4$	**0.625***	**0.625***	**0.000***	**0.750***
+img	0.500*	**0.625***	**0.000***	0.500*
+morph	0.375*	0.250*	**0.000***	0.625*
+both	0.500*	0.000*	**0.000***	0.500*
$\mathcal{F}_5$	**0.875***	**0.625***	**0.125***	**0.875***
+img	0.750*	**0.625***	**0.125***	0.750*
+morph	**0.875***	0.375*	0.000*	0.750*
+both	**0.875***	0.125*	0.000*	0.750*
$\mathcal{F}_6$	**0.750***	**0.500***	0.250*	**0.875***
+img	**0.750***	**0.500***	**0.375***	0.750*
+morph	0.500*	**0.500***	0.125*	0.750*
+both	0.625*	0.375*	0.000*	0.625*
Avg	**0.719***	**0.594***	0.094*	**0.781***
+img	0.656*	0.563*	**0.156***	0.625*
+morph	0.563*	0.313*	0.031*	0.594*
+both	0.625*	0.156*	0.031*	0.563*
Relapse (AUROC)				
$\mathcal{F}_3$	**0.800**	0.800	**0.667**	0.600
+img	**0.800**	0.800	**0.667**	0.700
+morph	0.633	**0.833**	0.500	**0.800**
+both	0.733	**0.833**	0.500	**0.800**
$\mathcal{F}_4$	0.600	**0.800**	**0.567**	0.500
+img	**0.800**	0.700	0.500	0.625
+morph	**0.800**	0.733	0.500	**0.708**
+both	0.633	0.733	0.500	0.583
$\mathcal{F}_5$	**0.600**	**0.800**	0.567	**0.600**
+img	**0.600**	0.600	**0.733**	**0.600**
+morph	0.433	0.733	0.500	0.500
+both	**0.600**	0.533	0.400	**0.600**
$\mathcal{F}_6$	0.700	**0.900**	0.567	0.500
+img	**0.800**	0.800	**0.733**	0.600
+morph	0.700	**0.900**	0.567	**0.700**
+both	**0.800**	0.800	0.567	0.500
$\mathcal{F}_7$	**0.500**	**0.500**	0.500	**0.500**
+img	**0.500**	**0.500**	0.500	**0.500**
+morph	**0.500**	**0.500**	**0.600**	**0.500**
+both	**0.500**	**0.500**	0.500	**0.500**
Avg	0.640	**0.760**	0.573	0.540
+img	**0.700**	0.680	**0.627**	0.605
+morph	0.613	0.740	0.533	**0.642**
+both	0.653	0.680	0.493	0.597

Table 6 Information gain from bone marrow imaging at every clinical feature level, external survival tasks (merged external cohort). Columns as in Table 5. C-index for EFS and OS. Base rows are re-evaluated on the morphology-available patient subset, so they can differ slightly from Table 3. Pooling the two source hospitals gives the morphology-available subset EFS/OS events, so survival is evaluable across the whole panel (no —). **Bold**: best variant within each level block (per column; ties all bolded). The pooled external fair-comparison subset is small (n=13); indicative only.

	Gm	Gma	Iv	Qw
EFS (C-index)				
$\mathcal{F}_3$	0.885	0.846	0.692	**0.923**
+img	0.846	0.744	0.654	0.795
+morph	**0.897**	**0.910**	**0.833**	0.744
+both	0.795	0.872	0.603	0.782
$\mathcal{F}_4$	0.833	0.846	0.692	**0.923**
+img	**0.846**	0.744	0.603	0.782
+morph	0.808	**0.910**	**0.833**	0.744
+both	0.667	0.872	0.641	0.782
$\mathcal{F}_5$	0.872	**0.897**	0.872	**0.577**
+img	**0.910**	0.885	0.692	0.500
+morph	0.859	0.872	**0.885**	**0.577**
+both	0.769	0.872	0.731	0.500
$\mathcal{F}_6$	**0.885**	0.782	0.859	0.680
+img	0.846	0.821	0.756	0.731
+morph	0.808	**0.859**	**0.897**	**0.859**
+both	0.859	0.782	0.795	0.641
$\mathcal{F}_7$	**0.821**	0.756	**0.872**	0.680
+img	**0.821**	0.718	0.692	0.551
+morph	0.782	**0.782**	0.859	**0.692**
+both	0.769	0.718	0.718	0.590
Avg	**0.859**	0.826	0.797	**0.756**
+img	0.854	0.782	0.679	0.672
+morph	0.831	**0.867**	**0.862**	0.723
+both	0.772	0.823	0.697	0.659
OS (C-index)				
$\mathcal{F}_3$	**0.850**	**0.883**	0.600	0.650
+img	0.817	0.750	0.600	0.750
+morph	0.817	0.850	0.600	**0.833**
+both	0.817	0.833	**0.700**	0.700
$\mathcal{F}_4$	0.817	**0.883**	**0.600**	0.650
+img	0.817	0.750	0.550	0.750
+morph	**0.900**	0.850	**0.600**	**0.833**
+both	0.833	0.833	**0.600**	0.700
$\mathcal{F}_5$	**0.933**	**0.933**	0.950	**0.850**
+img	0.900	0.783	0.950	0.500
+morph	0.800	0.900	0.867	0.800
+both	0.817	0.833	**0.983**	0.500
$\mathcal{F}_6$	0.800	0.700	0.850	0.817
+img	0.833	0.750	0.817	0.567
+morph	0.817	0.700	0.833	**0.850**
+both	**0.850**	**0.800**	**0.917**	0.567
$\mathcal{F}_7$	0.817	0.700	0.833	0.667
+img	**0.850**	0.617	**0.933**	0.650
+morph	**0.850**	0.700	0.833	**0.900**
+both	0.800	**0.733**	0.917	0.717
Avg	**0.843**	**0.820**	0.767	0.727
+img	**0.843**	0.730	0.770	0.643
+morph	0.837	0.800	0.747	**0.843**
+both	0.823	0.807	**0.823**	0.637

Table 7 Method comparison on the internal cohort, per temporal split. Rows are methods: LLM zero-shot (Gemini, best feature level, evaluated on each split's test set); decision tree (DT, a survival tree for EFS/OS) and random forest (RF, a random survival forest RSF for EFS/OS), our Phase 2 models. Classification: AUROC; survival (EFS, OS): C-index. **Bold** marks the best method in each split. Test sets by enrolment year: Split 1 (2017–19, n=201), Split 2 (2018–19, n=114), Split 3 (2019, n=17). —: no OS events in the 2019 test set. Full per-feature-level detail in Table 8.

Method	**Split 1** n=201	**Split 2** n=114	**Split 3** n=17
Final Risk (AUROC)			
LLM (Gemini)	0.860	0.856	**1.000**
DT	0.893	0.949	0.979
RF	**0.996**	**0.998**	**1.000**
D19 MRD (AUROC)			
LLM (Gemini)	0.651	0.745	0.560
DT	0.555	0.743	**0.893**
RF	**0.696**	**0.853**	0.833
D46 MRD (AUROC)			
LLM (Gemini)	**0.614**	0.632	0.560
DT	0.545	**0.671**	0.464
RF	0.597	0.636	**0.655**
Relapse (AUROC)			
LLM (Gemini)	**0.593**	0.589	**0.786**
DT	0.559	0.529	0.500
RF	0.591	**0.630**	0.500
EFS (C-index)			
LLM (Gemini)	0.661	0.674	**0.772**
DT	0.674	0.718	0.691
RSF	**0.684**	**0.746**	0.706
OS (C-index)			
LLM (Gemini)	0.682	**0.665**	—
DT	0.562	0.562	—
RSF	**0.688**	0.628	—

OS, RSF C-index was 0.629 ($p = 0.489$; not significant), reflecting the low event rate (8.0%) and limited statistical power.

External validation (merged cohort, OOD). With all 301 internal patients as training, RF and DT retained strong Final Risk discrimination on the pooled external cohort (best DT AUROC 0.968, RF 0.991 at $\mathcal{F}_7$; Table 9), confirming transfer to an independent patient population. Pooling the two source hospitals also resolves the single-class collapse seen when they were evaluated separately: Relapse and D19MRD are now two-class across the full external cohort and AUROC-evaluable — best

Table 8 Phase 2 internal results across all feature levels and the three temporal splits (full detail behind the summary in Table 7). Classification: DT / RF AUROC; survival (EFS, OS): survival-tree (DT) / RSF C-index. **Bold** marks the best feature level in each column. Splits by enrolment year: S1 train ≤16/test 17–19; S2 ≤17/18–19; S3 ≤18/19. —: not evaluable (D19/D46MRD are inputs at $\mathcal{F}_7$; no OS events in the 2019 test set). $\mathcal{F}_3^{+\mathrm{m}}$: $\mathcal{F}_3$ plus the quantitative morphology features (Section 2.3).

	Split 1		**Split 2**		**Split 3**	
Lv.	DT	RF	DT	RF	DT	RF
Final Risk (DT / RF AUROC)						
$\mathcal{F}_1$	0.759	0.842	0.810	0.871	0.799	0.806
$\mathcal{F}_2$	0.759	0.820	0.810	0.830	0.778	0.847
$\mathcal{F}_3$	0.711	0.849	0.782	0.872	0.771	0.743
$\mathcal{F}_3^{+\mathrm{m}}$	0.731	0.878	0.784	0.875	0.771	0.750
$\mathcal{F}_4$	0.711	0.852	0.757	0.847	0.771	0.799
$\mathcal{F}_5$	**0.893**	0.897	0.909	0.943	0.736	0.847
$\mathcal{F}_6$	0.893	0.922	0.911	0.951	0.736	0.882
$\mathcal{F}_7$	0.891	**0.996**	**0.949**	**0.998**	**0.979**	**1.000**
D19 MRD (DT / RF AUROC)						
$\mathcal{F}_1$	0.538	0.512	0.483	0.425	0.393	0.643
$\mathcal{F}_2$	0.497	0.487	0.495	0.495	0.464	0.667
$\mathcal{F}_3$	**0.555**	0.629	0.687	0.767	**0.893**	**0.833**
$\mathcal{F}_3^{+\mathrm{m}}$	0.555	**0.696**	**0.743**	0.807	0.821	0.774
$\mathcal{F}_4$	0.555	0.641	0.725	**0.853**	0.893	0.786
$\mathcal{F}_5$	0.555	0.668	0.687	0.764	0.750	0.738
$\mathcal{F}_6$	0.555	0.643	0.687	0.840	0.845	0.833
$\mathcal{F}_7$	—	—	—	—	—	—
D46 MRD (DT / RF AUROC)						
$\mathcal{F}_1$	0.364	0.534	0.405	0.516	**0.464**	0.619
$\mathcal{F}_2$	0.519	0.396	0.405	0.533	0.321	**0.655**
$\mathcal{F}_3$	**0.545**	0.564	0.519	0.605	0.369	0.309
$\mathcal{F}_3^{+\mathrm{m}}$	0.477	0.580	0.514	0.525	0.417	0.524
$\mathcal{F}_4$	0.531	0.528	**0.671**	**0.636**	0.369	0.417
$\mathcal{F}_5$	0.545	0.555	0.548	0.550	0.369	0.167
$\mathcal{F}_6$	0.545	**0.597**	0.548	0.613	0.369	0.464
$\mathcal{F}_7$	—	—	—	—	—	—
Relapse (DT / RF AUROC)						
$\mathcal{F}_1$	0.480	0.439	**0.529**	0.465	0.274	0.476
$\mathcal{F}_2$	0.550	0.546	0.483	0.526	**0.500**	0.357
$\mathcal{F}_3$	0.528	0.483	0.411	0.485	0.286	0.274
$\mathcal{F}_3^{+\mathrm{m}}$	**0.559**	0.516	0.411	0.561	0.333	0.202
$\mathcal{F}_4$	0.540	0.564	0.411	0.471	0.286	0.036
$\mathcal{F}_5$	0.527	0.517	0.411	0.624	0.286	0.286
$\mathcal{F}_6$	0.528	0.547	0.411	**0.630**	0.488	**0.500**
$\mathcal{F}_7$	0.529	**0.591**	0.411	0.597	0.488	0.429
EFS (DT / RSF C-index)						
$\mathcal{F}_1$	0.468	0.532	0.443	0.610	**0.691**	**0.706**
$\mathcal{F}_2$	0.482	0.516	0.501	0.597	0.529	0.559
$\mathcal{F}_3$	0.527	0.542	0.477	0.552	0.390	0.500
$\mathcal{F}_3^{+\mathrm{m}}$	0.557	0.530	0.547	0.516	0.507	0.397
$\mathcal{F}_4$	0.546	0.520	0.482	0.533	0.426	0.485
$\mathcal{F}_5$	0.494	0.511	0.490	0.572	0.390	0.588
$\mathcal{F}_6$	0.497	0.560	0.485	0.592	0.390	0.662
$\mathcal{F}_7$	**0.674**	**0.684**	**0.718**	**0.746**	0.640	0.691
OS (DT / RSF C-index)						
$\mathcal{F}_1$	0.541	**0.688**	0.531	0.497	—	—
$\mathcal{F}_2$	0.467	0.651	0.541	0.540	—	—
$\mathcal{F}_3$	0.512	0.556	0.440	0.515	—	—
$\mathcal{F}_3^{+\mathrm{m}}$	0.505	0.536	0.402	0.415	—	—
$\mathcal{F}_4$	0.498	0.590	0.472	0.529	—	—
$\mathcal{F}_5$	0.492	0.586	0.445	0.622	—	—
$\mathcal{F}_6$	**0.562**	0.621	**0.562**	0.620	—	—
$\mathcal{F}_7$	0.562	0.607	0.505	**0.628**	—	—

Relapse 0.583 (DT) / 0.528 (RF) and best D19MRD 0.628 (DT) / 0.635 (RF; Table 10) — while D46MRD still has only one positive case after pooling and is reported as accuracy (DT 0.732, RF 0.976). For survival, the pooled cohort (24 EFS events, 7 deaths) gave best RSF C-indices of 0.608 (EFS, $\mathcal{F}_7$) and 0.718 (OS, $\mathcal{F}_1/\mathcal{F}_2$), with DT reaching 0.603 (EFS, $\mathcal{F}_1$) and 0.710 (OS, $\mathcal{F}_7$).

2.5 Cosine RAG belief revision

For each test patient i, the $K = 5$ most similar training cases were retrieved by cosine similarity in the standardised feature space and used to compute a similarity-weighted RAG evidence probability p_{rag} (Eq. 4). The revised probability is $p_{\text{rev}} = \alpha\, p_0 + (1-\alpha)\, p_{\text{rag}}$, where $\alpha = 1 - (1-\alpha_{\text{base}})\sqrt{\bar{s}}$ and $\alpha_{\text{base}} = 0.6$.

Internal cohort. Table 11 reports AUROC before and after belief revision. Belief revision improved performance on Final Risk (Split 2: +0.002), D19MRD (Split 2: +0.012; Split 3: +0.024), and most substantially on D46MRD (Split 2: +0.046; Split 3: 0.655 → 0.881, $\Delta = +0.226$). The large D46MRD gain on Split 3 reflects high retrieval quality: with 283 training cases, the five nearest neighbours share strong phenotypic similarity (mean cosine $\bar{s} \approx 0.91$) and concordant D46MRD outcomes. Belief revision degraded Relapse performance (Split 2: 0.630 → 0.588, $\Delta = -0.042$): because relapse prediction is intrinsically noisy, retrieved neighbours have mixed outcomes, and RAG evidence acts as a noise source rather than a corrective signal.

External (pooled across sites). On the pooled external cohort, belief revision improved Final Risk (0.991 → 1.000, +0.009). D19MRD AUROCs degraded (−0.064) because the external set has very few D19MRD-positive patients ($n_{\text{pos}} = 4$), making retrieved neighbour outcomes uninformative. D46MRD belief revision could not be computed ($n_{\text{pos}} = 1$). These results confirm the internal boundary condition: belief revision benefits tasks where the feature space clusters meaningfully by label, but is counter-productive when class representation is too sparse to yield informative neighbours.

Survival belief revision. For EFS and OS, belief revision was applied to normalised RSF risk scores using event indicators as binary outcomes. Effects were near-neutral for EFS (mean ΔC-index $= -0.003$ across splits) and slightly negative for OS (mean ΔC-index $= -0.007$), where the low event rate (8%) renders most retrieved neighbours censored and the RAG evidence is uninformative.

2.6 LLM clinical narratives and qualitative evaluation

After belief revision, an LLM prompt was constructed for each test patient containing: (i) the patient's serialised feature vector; (ii) the RF-predicted label, revised probability p_{rev}, and the human-readable DT decision path from root to leaf; (iii) the six patient-salient features ranked by importance$_j \times |z_{ij}|$; and (iv) the structured profiles and outcomes of the top-5 retrieved historical cases. Gemini-3.1-Flash-Lite produced 3–5-sentence narratives for each prediction; all narratives from the $\mathcal{F}_7$/Split 3 experiment ($n = 17$ test patients) are available for physician review.

Two representative cases from the $\mathcal{F}_7$/Split 3 experiment illustrate the range of narrative quality (Fig. 6). In Case 1 (correctly classified IR, persistent MRD), the

Table 9 Phase 2 out-of-distribution results on the pooled external cohort, across all feature levels (train = all 301 internal patients; test = the two source hospitals pooled into a single external cohort, n=41). Classification: DT / RF AUROC; * = accuracy (D46MRD has only one positive case after pooling, so accuracy is reported instead). Survival (EFS, OS): survival-tree (DT) / random-survival-forest (RSF) C-index. **Bold** marks the best feature level in each column. —: not evaluable (D19/D46MRD are inputs at $\mathcal{F}_7$). Pooled external cohort; small, indicative only. $\mathcal{F}_3^{+\mathrm{m}}$: $\mathcal{F}_3$ plus the quantitative morphology features (Section 2.3).

Lv.	DT	RF
Final Risk (DT / RF)		
$\mathcal{F}_1$	0.805	0.878
$\mathcal{F}_2$	0.827	0.863
$\mathcal{F}_3$	0.779	0.843
$\mathcal{F}_3^{+\mathrm{m}}$	0.779	0.799
$\mathcal{F}_4$	0.778	0.750
$\mathcal{F}_5$	0.912	0.938
$\mathcal{F}_6$	0.912	0.934
$\mathcal{F}_7$	**0.968**	**0.991**
D19 MRD (DT / RF)		
$\mathcal{F}_1$	**0.628**	0.591
$\mathcal{F}_2$	0.466	**0.635**
$\mathcal{F}_3$	0.382	0.335
$\mathcal{F}_3^{+\mathrm{m}}$	0.264	0.294
$\mathcal{F}_4$	0.382	0.341
$\mathcal{F}_5$	0.392	0.307
$\mathcal{F}_6$	0.392	0.453
$\mathcal{F}_7$	—	—
D46 MRD (DT / RF)		
$\mathcal{F}_1$	**0.732***	**0.976***
$\mathcal{F}_2$	0.561*	0.976*
$\mathcal{F}_3$	0.488*	0.976*
$\mathcal{F}_3^{+\mathrm{m}}$	0.342*	0.976*
$\mathcal{F}_4$	0.244*	0.976*
$\mathcal{F}_5$	0.488*	0.976*
$\mathcal{F}_6$	0.488*	0.976*
$\mathcal{F}_7$	—	—
Relapse (DT / RF)		
$\mathcal{F}_1$	**0.583**	**0.528**
$\mathcal{F}_2$	0.397	0.494
$\mathcal{F}_3$	0.333	0.300
$\mathcal{F}_3^{+\mathrm{m}}$	0.447	0.390
$\mathcal{F}_4$	0.525	0.272
$\mathcal{F}_5$	0.333	0.377
$\mathcal{F}_6$	0.219	0.267
$\mathcal{F}_7$	0.250	0.383
EFS (DT / RSF C-index)		
$\mathcal{F}_1$	**0.603**	0.567
$\mathcal{F}_2$	0.552	0.572
$\mathcal{F}_3$	0.507	0.434
$\mathcal{F}_3^{+\mathrm{m}}$	0.576	0.439
$\mathcal{F}_4$	0.509	0.475
$\mathcal{F}_5$	0.419	0.444
$\mathcal{F}_6$	0.419	0.439
$\mathcal{F}_7$	0.490	**0.608**
OS (DT / RSF C-index)		
$\mathcal{F}_1$	0.619	**0.718**
$\mathcal{F}_2$	0.657	0.718
$\mathcal{F}_3$	0.571	0.468
$\mathcal{F}_3^{+\mathrm{m}}$	0.587	0.659
$\mathcal{F}_4$	0.702	0.512
$\mathcal{F}_5$	0.647	0.556
$\mathcal{F}_6$	0.647	0.512
$\mathcal{F}_7$	**0.710**	0.643

top-5 retrieved neighbours all share elevated MRD and cytogenetic abnormalities, and the narrative coherently attributes the high-risk classification to MRD non-clearance corroborated by historical precedent. In Case 2 (borderline case), the narrative highlights conflicting signals between MRD clearance and cytogenetic risk, suggesting next-generation sequencing as a way to resolve ambiguity.

Quantitative physician evaluation of narrative quality is planned and will follow the protocol in Table 12. Evaluation will cover: factual consistency with the patient's features, relevance of historical case comparisons, clinical plausibility of the interpretation, and overall helpfulness for clinical decision-making.

3 Discussion

This study presents a temporally-structured multimodal framework evaluated jointly across six ALL prediction targets—four classification tasks and two survival endpoints—and validated at an independent external centre. Several findings warrant discussion.

MRD as the dominant signal across all targets. The largest single performance gain in Phase 2—DT AUROC for Final Risk from 0.912 ($\mathcal{F}_6$, cytogenetics) to 0.949 ($\mathcal{F}_7$, plus D19/D46 MRD)—quantitatively confirms the central role of MRD in clinical risk stratification [11]. The same pattern holds for LLM EFS C-index, which peaked at feature levels including MRD (Gemma: 0.716 at $\mathcal{F}_7$), and for Phase 2 RSF EFS (C-index 0.746, $p = 0.010$). These convergent results from four LLMs, decision trees, and a survival forest provide multi-method evidence that residual disease burden at Day 46 is the strongest prognostic signal accessible before relapse.

LLM zero-shot as a competitive baseline. Phase 1 Final Risk AUROCs of 0.848–0.891 without labelled training data are competitive with Phase 2 DT at $\mathcal{F}_6$ (0.912) and only modestly below the Phase 2 DT peak (0.949 at $\mathcal{F}_7$). This indicates that clinical knowledge encoded during LLM pretraining includes a meaningful understanding of paediatric ALL risk factors, and that LLMs may serve as strong priors when historical case databases are small or absent—for example, at the time a new treatment protocol is introduced. On the merged external cohort, LLM Final Risk AUROCs at the best feature level ($\mathcal{F}_7$) ranged 0.750–0.925 across models, surpassing the internal zero-shot values for some models, likely due to differences in class distribution (larger IR/HR proportion in the external cohort) rather than genuine improvement in discriminative ability.

Imaging information: morphology text, not raw pixels. The multimodal comparison (Section 2.3) shows an asymmetry at diagnosis: the structured morphology profile improved zero-shot relapse prediction for all four LLMs (ΔAUROC +0.003 to +0.040), whereas raw smear images improved Final Risk for three of four models only marginally and degraded relapse prediction for three of four. This suggests that current multimodal LLMs extract prognostic morphology more reliably from a quantitative, pipeline-derived text profile than from raw microscopy, supporting automated morphometry as the practical route for injecting imaging information into clinical risk models. The completed comparisons across all five feature levels reinforce this asymmetry: the morphology profile improved relapse prediction for a majority of models at

three of five levels and EFS in 15 of 20 model–level comparisons, and even at $\mathcal{F}_7$—with MRD already among the inputs—it improved Final Risk for all four models (+0.005 to +0.102), whereas raw images degraded Final Risk there for all four. The effects remain small and unadjusted, but they indicate that the morphometric text profile retains additive prognostic value even after molecular and MRD data arrive, while raw pixels do not.

External validation: generalisation and limitations. Phase 2 RF Final Risk AUROC of 0.991 on the merged external cohort (versus 0.998 internally) demonstrates strong generalisation across independent institutions. However, D19MRD AUROC dropped from 0.807 (internal) to 0.635 (external), indicating feature-distribution shift that partly limits generalisation for this target. The single D46MRD-positive patient remaining after pooling ($n = 1$ of 29 labelled patients) precluded belief revision and produced unstable AUC estimates, highlighting the need for larger external cohorts with balanced outcome distributions. The pooled external cohort remains small (41 patients from two hospitals of 20 and 21) and includes a relapse-enriched referral subcohort (see Limitations), so some targets remain event-sparse even after pooling (e.g., D46MRD, $n = 1$ positive); larger, prospectively balanced external cohorts remain necessary for a definitive test of cross-institution robustness.

Integrating six targets in a unified framework. Treating EFS and OS as targets 5–6 of the same feature hierarchy—rather than studying them in a separate survival analysis—reveals a coherent picture: the same feature level ($\mathcal{F}_7$) that maximises Final Risk classification also maximises EFS survival discrimination (RSF C-index 0.746), while Relapse and OS remain challenging regardless of feature set. This joint view avoids the selective reporting that can arise when survival and classification tasks are presented in separate papers or using different patient subsets.

Boundary conditions for RAG belief revision. Belief revision delivers quantitative gains specifically when the feature space clusters meaningfully by label. For D46MRD at Split 3 (mean similarity $\bar{s} \approx 0.91$), the $\Delta + 0.226$ AUROC gain is substantial and clinically relevant. For Relapse ($\Delta - 0.042$ internally, +0.016 externally with very low base AUROC), and for OS (mean ΔC-index $= -0.007$), retrieved neighbours are too heterogeneous to provide reliable evidence. A practical deployment criterion would be to apply belief revision only when inter-neighbour outcome agreement exceeds a calibrated threshold—for example, Fleiss $\kappa > 0.4$ over the top-K retrieved labels.

Relapse remains an open problem. Relapse prediction was intractable for all Phase 1 and Phase 2 models (best DT AUROC 0.529, LLM best 0.643), consistent with the broader literature [4, 8]. Our feature set does not include genomic or epigenomic data (e.g., copy-number variation, DNA methylation), which have shown modest incremental utility [6]. Future multi-omics integration within the temporal feature hierarchy may improve relapse prediction.

Limitations. The internal cohort is a single-institution retrospective study, and the external cohort—though pooled across two hospitals—remains small; multi-centre prospective validation at greater scale is required before clinical deployment. The internal cohort ($n = 301$) limits statistical power for OS (24 events) and Split 3 ($n_{\text{test}} = 17$). The imaging comparison is complete at all feature levels internally,

but its external fair-comparison subsets remain small (classification $n = 8$, survival $n = 13$); pooling now makes external survival evaluable across the full feature range, but the imaging findings should still be read primarily as internal-cohort evidence. The external cohort pools two hospitals, one of which (Fudan) is a relapse-enriched referral cohort in which all patients relapsed; the pooled external relapse rate is therefore far above that of an unselected ALL population, so external metrics—particularly for relapse—are exploratory rather than representative estimates; moreover, the complete-label subset used for the Phase 1 classification metrics (n=29) is more skewed still (23 of 29 relapsed, 79%), because joint-label completeness retained all relapse-positive patients. Phase 1 LLMs were evaluated zero-shot; fine-tuning on labelled clinical data may yield further improvements. Physician evaluation of LLM narratives (Table 12) is ongoing and will determine whether the narratives are clinically actionable beyond proof-of-concept case studies.

4 Methods

4.1 Datasets and ethics

The internal training cohort comprised $n = 301$ patients diagnosed with ALL at [HOSPITAL NAME] between 2015 and 2019. All patients were treated with risk-stratified chemotherapy protocols consistent with contemporary Chinese paediatric oncology guidelines. Ethics approval was obtained from [ETHICS NUMBER]; all data were de-identified prior to analysis. Each patient record includes structured clinical data (demographics, haematological indices, immunophenotype, 51-marker flow cytometry panel, cytogenetics, MRD values) and bone marrow aspirate microscopy images (Giemsa-stained, ∼20 fields per patient).

The external validation cohort ($n = 41$) was supplied as a single merged file spanning two source hospitals, distinguished by the recorded hospital field: Lianyungang First People's Hospital (20 patients, 2019–2023) and Children's Hospital of Fudan University, Shanghai (21 patients, 2017–2021, a relapse-enriched referral cohort in which all patients relapsed); ethics approval: [ETHICS NUMBERS, external cohort]. We pool the two hospitals and evaluate the external cohort as a single group throughout (Section 4.7). No external data were used in any model training.

4.2 Temporally-structured feature hierarchy

Feature sets are temporally ordered and, except for the $\mathcal{F}_4$ branch, form a strictly nested sequence $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3 \subseteq \mathcal{F}_5 \subseteq \mathcal{F}_6 \subseteq \mathcal{F}_7$ ($\mathcal{F}_4$ branches from $\mathcal{F}_3$; Table 14), where each level contains exactly the features available at the corresponding clinical timepoint. Boundaries: $\mathcal{F}_1$ (Day 0): WBC count and age; $\mathcal{F}_2$ (Day 1): adds blast percentage; $\mathcal{F}_3$ (Day 1): adds immunophenotype and 51 flow cytometry markers; $\mathcal{F}_4$ (Day 1): adds non-risk clinical factors; $\mathcal{F}_5$ (Day 5): adds CNS status, Day 5 blast, and Initial Risk; $\mathcal{F}_6$ (Day 7–10): adds Fusion gene, Karyotype, FISH; $\mathcal{F}_7$ (Day 46): adds D19MRD and D46MRD. The complete variable list for every level is given in Table 14. All six prediction targets are evaluated on the same feature hierarchy.

For Phase 2, categorical features were one-hot encoded and numerical features z-score normalised using training-split statistics only:

$$z_{ij}^{(\text{test})} = \frac{x_{ij}^{(\text{test})} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \tag{1}$$

with missing values imputed by training-split medians.

4.3 Bone marrow morphology pipeline

Stage 1: Cell segmentation. SAM 3.1 [13] was applied with the text prompt *"complete dark-stained nucleated cell"* after colour-based pre-filtering (HSV hue 120–160°) to restrict sampling to haematoxylin-stained regions.

Stage 2: Morphological feature extraction. For each retained cell, ∼37 features were computed: geometric descriptors (area, perimeter, circularity, eccentricity, aspect ratio), textural descriptors (GLCM contrast and correlation), nucleus-to-cytoplasm (N/C) ratio, nucleoli count, and vacuole percentage.

Stage 3: Cell-type clustering. BiomedCLIP [14] was applied in a zero-shot clustering mode with $k = 7$ prototype classes (blast, granulocyte, megakaryocyte, monocyte, plasma cell, red cell, other), with Hungarian algorithm assignment.

Stage 4: Per-patient aggregation. Cell-level features were aggregated into per-patient morphology statistics (cell-type proportions and summary descriptors) that augment the tabular Phase 2 models; the same profile is serialised as the +morph text input in the multimodal comparison (Section 2.3).

4.4 Phase 1: LLM zero-shot inference

For classification tasks, each patient's features were serialised into a key–value list and presented with a task-specific system prompt requesting a bracketed label (e.g., `[[IR]]`/`[[LR]]`). For survival risk scoring, the same serialised features were presented with a prompt requesting a continuous event probability (0.0–1.0). All experiments used temperature = 0 (greedy decoding). Per-patient outputs were cached; the cohort was scored for every combination of feature level (×7, plus the imaging variants of Section 2.3), task (×6), and model (×4).

4.5 Phase 2: Decision tree and random forest

A decision tree (DT; `max_depth`=4, `class_weight`='balanced', `criterion`='gini', `random_state`=42) and a random forest (RF; $n_{\text{estimators}} = 100$, `class_weight`='balanced') were trained for each feature level and temporal split. Three temporal splits defined by enrolment year threshold $\mathcal{T}_s$ (Eq. 2) ensure strict temporal non-leakage.

$$\mathcal{D}_s^{\text{train}} = \{i : \tau_i \leq \mathcal{T}_s\}, \quad \mathcal{D}_s^{\text{test}} = \{i : \tau_i > \mathcal{T}_s\}. \tag{2}$$

Primary metric: AUROC; secondary: F1-macro and balanced accuracy.

4.6 Phase 2: Random survival forest

A random survival forest (RSF; [17]; $n_{\text{estimators}} = 100$, `random_state`=42) was trained on the same temporal splits as the classification experiments; a single survival decision tree (survival-tree analogue of the classification DT) was trained under the same protocol as its tree-based counterpart. EFS was defined as time from diagnosis to first of: relapse, induction failure, or death; OS was time to death. Censoring was at last follow-up. Evaluation used Harrell's C-index (Eq. 6), IPCW time-dependent AUC (Eq. 7) at $t^* \in \{12, 24, 36\}$ months, and the log-rank test on Kaplan-Meier curves dichotomised at the median risk score.

4.7 External OOD evaluation

For external validation, all $n = 301$ internal patients served as the training set and external patients served as the test set; no site-specific fine-tuning was performed. DT and RF were trained on the full internal set using the same preprocessing pipeline (Eq. 1). RSF was trained identically. For Phase 1 LLM evaluation, no training was involved; the same zero-shot prompts were applied to external patient feature vectors.

4.8 RAG belief revision and LLM narration

For test patient i, the $K = 5$ most similar training cases were retrieved by cosine similarity in the standardised feature space:

$$\mathcal{N}(i) = \underset{j \in \mathcal{D}^{\text{train}}}{\arg\,\text{top-}K} \; \frac{\mathbf{z}_i \cdot \mathbf{z}_j}{\|\mathbf{z}_i\|_2 \|\mathbf{z}_j\|_2}, \quad K = 5. \tag{3}$$

The RAG evidence probability is

$$p_{\text{rag}} = \frac{\sum_{j \in \mathcal{N}(i)} s_j \, y_j}{\sum_{j \in \mathcal{N}(i)} s_j}, \tag{4}$$

where s_j is the cosine similarity to neighbour j and $y_j \in \{0, 1\}$ is the binary task outcome (or event indicator for survival). The revised probability is

$$p_{\text{rev}} = \alpha \, p_0 + (1 - \alpha) \, p_{\text{rag}}, \qquad \alpha = 1 - (1 - \alpha_{\text{base}}) \sqrt{\bar{s}}, \tag{5}$$

where $\alpha_{\text{base}} = 0.6$ and $\bar{s} = \frac{1}{K} \sum_{j \in \mathcal{N}(i)} s_j$. For survival, p_0 is the min-max normalised RSF risk score.

After belief revision, an LLM prompt containing the patient's features, DT decision path, top-6 salient features (ranked by importance$_j \times |z_{ij}|$), and top-5 retrieved case profiles was submitted to Gemini-3.1-Flash-Lite to generate a 3–5 sentence clinical narrative.

4.9 Survival metrics

Harrell's concordance index:

$$\hat{C} = \frac{\sum_{i \neq j} \mathbf{1}[T_i < T_j]\, \mathbf{1}[\delta_i = 1]\, \mathbf{1}[\hat{r}_i > \hat{r}_j]}{\sum_{i \neq j} \mathbf{1}[T_i < T_j]\, \mathbf{1}[\delta_i = 1]}. \tag{6}$$

IPCW time-dependent AUC:

$$\widehat{\mathrm{AUC}}(t^*) = \frac{\sum_{i:\, T_i \leq t^*,\, \delta_i = 1} \sum_{j:\, T_j > t^*} \mathbf{1}[\hat{r}_i > \hat{r}_j]\, \hat{w}_i}{\sum_{i:\, T_i \leq t^*,\, \delta_i = 1} \sum_{j:\, T_j > t^*} \hat{w}_i}, \tag{7}$$

where $\hat{w}_i = 1/\hat{G}(T_i)^2$ and $\hat{G}(\cdot)$ is the Kaplan-Meier censoring estimator [16].

4.10 Statistical analysis

AUROC was computed with `scikit-learn` [18]; C-index and IPCW AUC with `scikit-survival` [19]. Bootstrap 95% confidence intervals (1000 resamples) are reported for Split 3 ($n_{\text{test}} = 17$). No multiple-testing correction was applied given the exploratory nature of this study; all reported p-values are two-sided.

Data availability

The clinical dataset contains patient health information and cannot be made publicly available due to privacy regulations. De-identified data may be available through a data access agreement with [HOSPITAL NAME]; requests to the corresponding author.

Code availability

Analysis code, LLM prompts, and the bone marrow morphology pipeline will be made available at https://github.com/YanbeiJiang/ALL_prediction_NCS upon acceptance.

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Author contributions

[To be completed]

Competing interests

The authors declare no competing interests.

Fig. 1 Overview of the temporally-structured multimodal prediction framework. a, The three parallel timelines: feature acquisition (Day 0–46, $\mathcal{F}_1$–$\mathcal{F}_7$), prediction targets (six tasks: Final Risk, D19MRD, D46MRD, Relapse, EFS, OS), and the dynamic case database (patient enrolment year 2015–2019, temporal splits). **b,** Phase 1 architecture: four LLMs receive serialised patient features and output class labels or risk scores without task-specific training. **c,** Phase 2 architecture: a decision tree trained on historical cases generates predictions and decision paths; RAG retrieves the $K = 5$ most similar cases; LLM synthesises a clinical narrative. **d,** Evaluation scope: internal temporal splits (three splits) and external OOD validation (merged external cohort).

Fig. 2 Phase 1 feature-level ablation across six prediction targets. a, Final Risk AUROC vs feature level for four LLMs. **b,** Relapse AUROC. **c,** D19MRD AUROC. **d,** D46MRD AUROC. **e,** EFS C-index. **f,** OS C-index. Vertical dashed lines indicate clinical timepoints: Day 1 (flow cytometry), Day 5 (early response), Day 7–10 (cytogenetics), Day 46 (MRD, $\mathcal{F}_7$). Open symbols: internal cohort. Filled symbols: External cohort.

Fig. 3 Bone marrow morphology pipeline and its predictive contribution. a, Pipeline schematic: SAM cell segmentation → morphological feature extraction → BiomedCLIP cell-type clustering → per-patient morphology profile. **b,** Representative segmented cells with BiomedCLIP cell-type assignments. **c,** Predictive contribution of the morphology features in Phase 2: D19MRD DT/RF AUROC at $\mathcal{F}_3$ (no morphology) versus $\mathcal{F}_3^{+\text{morph}}$ (with morphology features).

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Fig. 4 Phase 2 temporal learning curves and classification results. a, Final Risk DT AUROC across feature levels for three temporal splits; shaded band: 95% bootstrap CI for Split 3 ($n_{\text{test}} = 17$). **b,** Training-set size vs AUROC for Final Risk and Relapse (DT and RF) at $\mathcal{F}_7$. **c,** External cohort RF AUROC comparison with internal Split 2 RF AUROC across all four classification tasks.

Fig. 5 Survival prediction: Kaplan-Meier curves and C-index summary. a, EFS Kaplan-Meier curves (high vs low risk, median split) for the best Phase 1 model (Gemma, $\mathcal{F}_7$, internal cohort). **b,** EFS KM curves for Phase 2 RSF (Split 2, $\mathcal{F}_7$, log-rank $p = 0.010$). **c,** OS KM curves for Phase 2 RSF on the external cohort (log-rank $p = 0.031$). **d,** C-index heatmap: all LLMs $\times$ all feature levels for EFS (internal, left; external cohort, right). Numbers at risk and 95% CI shown.

Fig. 6 RAG belief revision and representative case narratives. a, AUROC before vs after belief revision across four tasks and three splits; point size proportional to training-set size. Filled points: internal; open points: External cohort. **b–c,** Two representative cases from $\mathcal{F}_7$/Split 3. For each case: patient feature summary, DT decision path, top-3 retrieved similar historical cases with outcomes, and LLM-generated clinical narrative. Case b: correctly classified IR patient with persistent MRD and concordant historical neighbours. Case c: borderline case with conflicting MRD clearance vs cytogenetic risk signals.

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Table 10 Method comparison on the pooled external cohort (out-of-distribution; no temporal split; the two source hospitals pooled into a single cohort, n=41). Rows are methods: LLM zero-shot (Gemini, best feature level); decision tree (DT, a survival tree for EFS/OS) and random forest (RF, RSF for EFS/OS), trained on all 301 internal patients. Classification: AUROC (* = accuracy; D46MRD has only one positive case after pooling, so accuracy is reported instead); survival (EFS, OS): C-index. **Bold** marks the best method for each task. Pooled external cohort; small, indicative only. Full per-feature-level detail in Table 9.

Method	**External** n=41
Final Risk (AUROC)	
LLM (Gemini)	0.925
DT	0.968
RF	**0.991**
D19 MRD (AUROC)	
LLM (Gemini)	0.550
DT	0.628
RF	**0.635**
D46 MRD (accuracy)	
LLM (Gemini)	0.690*
DT	0.732*
RF	**0.976***
Relapse (AUROC)	
LLM (Gemini)	**0.717**
DT	0.583
RF	0.528
EFS (C-index)	
LLM (Gemini)	**0.715**
DT	0.603
RSF	0.608
OS (C-index)	
LLM (Gemini)	**0.810**
DT	0.710
RSF	0.718

Table 11 Cosine RAG belief revision: RF AUROC before (p_{RF}) and after (p_{rev}) belief revision ($K = 5$, $\alpha_{\mathrm{base}} = 0.6$), best experiment per task. Internal results are reported per temporal split; external results use the pooled external cohort, as the per-site external cohorts are too small or single-class for this analysis. **Bold**: largest gain. † insufficient positives; not computed.

Task	**Split**	p_{RF}	p_{rev}	Δ
Internal (temporal splits)				
Final Risk	Split 1	0.996	0.998	+0.002
Final Risk	Split 2	0.998	0.999	+0.002
Final Risk	Split 3	1.000	1.000	+0.000
D19MRD	Split 1	0.629	0.631	+0.002
D19MRD	Split 2	0.767	0.779	+0.012
D19MRD	Split 3	0.833	0.857	+0.024
D46MRD	Split 1	0.396	0.371	−0.025
D46MRD	Split 2	0.533	0.579	+0.046
D46MRD	Split 3	0.655	**0.881**	**+0.226**
Relapse	Split 1	0.547	0.546	−0.001
Relapse	Split 2	0.630	0.588	−0.042
Relapse	Split 3	0.500	0.476	−0.024
External (pooled cohort)				
Final Risk	—	0.991	1.000	+0.009
D19MRD	—	0.335	0.270	−0.064
D46MRD	—	† ($n_{\mathrm{pos}} = 1$)		
Relapse	—	0.267	0.283	+0.016

Table 12 Planned physician evaluation of LLM clinical narratives. Each narrative will be assessed on four dimensions by two independent clinicians using a 5-point Likert scale. Results to be filled following structured review.

Case ID	**Factual accuracy** (1–5)	**Retrieval relevance** (1–5)	**Clinical plausibility** (1–5)	**Decision helpfulness** (1–5)	**C**
Case 001	—	—	—	—	
Case 002	—	—	—	—	
Case 003	—	—	—	—	
⋮	⋮	⋮	⋮	⋮	
Case 017	—	—	—	—	
Mean (SD)	—	—	—	—	
Inter-rater κ	—	—	—	—	

Table 13 Feature set design aligned with the clinical data-acquisition timeline. All six prediction targets are evaluated at every feature level; † at $\mathcal{F}_7$, D19MRD and D46MRD are used as *input features*, so only Final Risk and Relapse are predicted. EFS/OS are evaluated at all levels using the same feature vector as classification tasks.

Level	Timepoint	Cumulative features added	Class. targets
$\mathcal{F}_1$	Day 0	WBC count, Age	4
$\mathcal{F}_2$	Day 1	+ Blast%	4
$\mathcal{F}_3$	Day 1	+ Immunophenotype + 51 flow cytometry markers	4
$\mathcal{F}_4$	Day 1	$\mathcal{F}_3$ + non-risk clinical factors	4
$\mathcal{F}_5$	Day 5	$\mathcal{F}_3$ + CNS status + Day 5 blast% + Initial Risk	4
$\mathcal{F}_6$	Day 7–10	$\mathcal{F}_5$ + Fusion gene + Karyotype + FISH	4
$\mathcal{F}_7$	Day 46	$\mathcal{F}_6$ + D19MRD + D46MRD (as inputs)	$2^{\dagger}$

Survival targets (EFS, OS) evaluated at all levels using the same feature vector.

Table 14 Complete variable specification for each feature level (Supplementary Table). Each level adds the listed variables to its base set; $\mathcal{F}_4$ branches from $\mathcal{F}_3$, while the main clinical chain is $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3 \subseteq \mathcal{F}_5 \subseteq \mathcal{F}_6 \subseteq \mathcal{F}_7$.

Level	Variables added (base set)
$\mathcal{F}_1$	WBC count; Age
$\mathcal{F}_2$	+ Blast%
$\mathcal{F}_3$	+ Immunophenotype (B/T lineage); 42 CD markers (CD1a, CD2, CD3, cCD3, CD4, CD5, CD7, CD8, CD9, CD10, CD11b, CD11c, CD13, CD14, CD15, CD19, CD20, CD22, cCD22, CD24, CD33, CD34, CD38, CD41, CD44, CD45, CD58, CD61, CD64, CD65, CD66c, CD71, CD73, cCD79a, CD86, CD90, CD97, CD99, CD117, CD123, CD133, CD200); 9 additional markers (GPA, cMPO, NG2, TdT, $c\mu$, sIgM, TCR$\alpha\beta$, TCR$\gamma\delta$, HLA-DR)
$\mathcal{F}_4$	(base $\mathcal{F}_3$) + non-risk clinical factors: chief-complaint category, pregnancy number, birth order, term/preterm status, delivery mode, birth weight, feeding type, admission weight, liver span, spleen span, haemoglobin, platelet count
$\mathcal{F}_5$	(base $\mathcal{F}_3$) + CNS status (CNSL); Day-5 peripheral blast%; Initial Risk
$\mathcal{F}_6$	(base $\mathcal{F}_5$) + Fusion gene; Karyotype; FISH
$\mathcal{F}_7$	(base $\mathcal{F}_6$) + D19MRD; D46MRD (as input features)
$\mathcal{F}_k^{+\text{img/morph/both}}$	(base $\mathcal{F}_k$, $k \in \{3, \ldots, 7\}$) + raw bone marrow smear images (+img), the structured morphology profile from the SAM/BiomedCLIP pipeline: per-patient blast-cell descriptors (area, perimeter, circularity, eccentricity, aspect ratio, nucleus-to-cytoplasm ratio, GLCM texture, chromatin fineness, nucleoli count/size, vacuole %, hand-mirror %) and cell-type proportions (+morph), or both (+both)